ANNA TAM LY

atl2@ualberta.ca | atn-ly.github.io

PROFESSIONAL SUMMARY

MSc student in Mathematical and Statistical Sciences (Modelling, Data, and Predictions) at the University of Alberta, graduating December 2026 and available for full-time analytical roles from January 2027; open to relocation. I apply strong quantitative training to real-world problems across three core areas:

- **Data Science:** Skilled in Python and R for data cleaning, analysis, and visualization; SQL (MySQL, SQLite) for database creation and management; Tableau for insight dashboards; and Git/GitHub for reproducible workflows. Built a computer-vision application that automated traffic-data reporting for a business improvement area.
- **Machine Learning:** Trained in supervised methods (regression, SVMs, tree-based models, neural networks) and unsupervised methods (clustering, PCA, factor analysis). Applied seven models to fill gaps in hourly soil-moisture records, validate predictions, and identify key environmental predictors for a government agricultural data tool.
- **Mathematical & Statistical Modelling:** Experienced in using MATLAB to translate real-world systems into models, analyze dynamics, optimize outcomes, and quantify uncertainty. Developed models of infectious-disease control and ecological adaptation for research labs specializing in artificial intelligence, public health, and genetics.

RELEVANT EXPERIENCE

AI & Business Data Analyst Intern **May 2026 - Aug 2026**

Old Strathcona Business Association, Edmonton, AB

- **Computer Vision App Development:** Built a Python app using YOLO and OpenCV to automatically count people and vehicles in video; tested it against hand counts and fixed tracking errors, raising accuracy from 51% to 89%.
- **ETL Pipeline Development:** Built a pipeline that extracted footage from 12 cameras, transformed detections into timestamped crossing counts using configurable counting lines, and loaded results into a SQLite database.
- **Data Analytics & Reporting:** Analyzed 50,000+ pedestrian counts by hour, day, location, weather, and event, producing visual reports to support OSBA's data-driven decision-making for events and district planning.
- **CRM & Database Modernization:** Evaluated CRM platforms to replace MembershipWorks, restructured 550+ business records into a relational database, and built n8n workflows to automate WordPress directory updates.
- **Survey Analysis & Community Insights:** Cleaned and analyzed 413 community-survey responses, comparing residents vs. visitors and coding 354 open-ended answers to identify infrastructure priorities on Whyte Avenue.

Data Science Graduate Student Researcher **May 2026 - Aug 2026**

University of Alberta, in collaboration with Alberta Agriculture and Irrigation, Edmonton, AB

- **Time-Series Data Preparation:** Built Python/Jupyter Notebook workflows to prepare raw Alberta Climate Information Service (ACIS) data, covering ~900,000 hourly records × four soil-depth variables across five stations.
- **Data Imputation & Predictive Modelling:** Developed a three-level gap-classification strategy and applied Random Forest, XGBoost, LightGBM, CatBoost, LSTM, GRU, and TCN to cut missing data from 3.7% to 0.02%.
- **Model Validation & Evaluation:** Simulated missing-data gaps by masking known observations, then validated predictions using MAE, RMSE, R^2, and bias, achieving $R^2 \geq 0.97$ for short-duration gaps.
- **Feature Importance Analysis:** Used permutation importance to determine which soil-moisture history, climate, and seasonal predictors contributed most to prediction accuracy across the four soil depths.
- **Research Communication:** Presented project methods, results, and limitations to university researchers and soil scientists, translating complex modelling concepts for audiences with different technical backgrounds.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, Matplotlib, seaborn), R (tidyverse), MATLAB, SQL, Stata

Data & Visualization: Relational databases (MySQL, SQLite, Microsoft Access), CRM platforms (MembershipWorks, Airtable, Baserow), dashboards and reporting (Tableau, Excel)

Modelling & Optimization: Statistical modelling (regression, GLMs, Bayesian methods, time series), mathematical modelling (differential equation, stochastic, Monte Carlo methods), optimization (deterministic, probabilistic/metaheuristic, multi-objective)

Machine Learning, AI & Automation: ML libraries (scikit-learn, PyTorch, TensorFlow, XGBoost), generative AI (LLMs, RAG, semantic/vector search), computer vision (YOLO, OpenCV), automation (ETL pipelines, n8n workflows, automated reporting)

Development: Version control (Git/GitHub), notebooks (Jupyter, Google Colab), app development (Streamlit, PySide6)

RESEARCH EXPERIENCE

Infectious Disease Modelling Undergraduate Student Researcher **Sep 2024 - Apr 2025**

AI & Mathematical Modelling (AIMM) Lab, Dalla Lana School of Public Health, University of Toronto, Toronto, ON

- **Compartmental Modelling:** Built and analyzed an ODE compartmental model of an mpox outbreak to examine disease transmission and evaluate potential control strategies.
- **Literature Synthesis:** Reviewed infectious-disease modelling literature to inform the project's research approach, assumptions, and interpretation of results.

Spatial Ecology Research Assistant **May 2024 - Aug 2024**

Fortin Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Data Entry:** Digitized 1,000+ historical tree records into Microsoft Excel for spatial ecology analyses.
- **Database Management:** Maintained a Microsoft Access database, ensuring data completeness and consistency.

Quantitative Genetics Undergraduate Student Researcher **May 2023 - Aug 2023**

Osmond Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Mathematical Modelling:** Investigated a PDE model describing adaptation to a moving environmental optimum.
- **Mathematical Analysis:** Derived expressions for mean fitness and the critical rate of environmental change.

Evolutionary Genetics Research Assistant **May 2022 - Nov 2022**

Agrawal Lab, Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON

- **Image Analysis & Data Collection:** Collected growth measurements for 200+ plants using ImageJ and Excel.
- **Statistical Analysis:** Performed t-tests to assess fitness differences across reproductive modes.

ADDITIONAL EXPERIENCE

Retail Specialist **Jul 2020 - Jul 2022**

Krispy Kreme Doughnut Factory, Scarborough, ON

- **Teamwork & Communication:** Trained new staff on doughnut and beverage preparation and communicated effectively with team members to keep service moving smoothly and ensure a positive customer experience.
- **Multitasking & Time Management:** Balanced register duties, order fulfillment, and customer requests simultaneously, consistently upholding speed and accuracy during high-volume shifts.
- **Attention to Detail & Reliability:** Processed transactions accurately and maintained clean, sanitary work areas in compliance with health and safety standards.

EDUCATION

University of Alberta, Master of Science (MSc) **Expected Dec 2026**

Mathematical and Statistical Sciences, Specialization in Modelling, Data, and Predictions

- GPA: 3.95/4.00

University of Toronto, Honours Bachelor of Science (HBSc) with High Distinction **Apr 2025**

Double Major in Mathematics and Quantitative Biology, Minor in Statistics

- GPA: 3.76/4.00

HONOURS AND AWARDS

Data Masters Challenge Olympiad **Apr 2026**

- Awarded second place in a 48-hour intercollegiate data science competition by analyzing 390 Olympic athlete records across 17 variables, engineering features, evaluating machine-learning models to predict medal success, and developing strategic recommendations for the United States and Canada ahead of the 2030 Winter Olympics.

University of Alberta Graduate Recruitment Scholarship **Feb 2025**

- Awarded $25,000 through a provincial government grant, fully funding tuition and living expenses for the MSc program, with fast-tracked admission in recognition of an outstanding application.

University of Toronto Ecology & Evolutionary Biology Undergraduate Student Research Award **May 2023**

- Awarded $7,500 to conduct a full-time summer research project in the Osmond Lab, modelling how populations adapt to a moving environmental optimum.

Dean's List Scholar **2021, 2022, 2023, 2025**

- Awarded to University of Toronto's Faculty of Arts & Science students who obtain a cumulative GPA of 3.50 or higher after completing their fifth, tenth, fifteenth, or twentieth degree credits.